Tangent & Normal: Formula Sheet

Tangent and normal

Slope:

$$m = \left.\frac{dy}{dx}\right|_{(x_1,y_1)}$$

Tangent: $y - y_1 = m(x - x_1)$

Normal: $y - y_1 = -\frac{1}{m}(x - x_1)$

Parametric slope:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Horizontal tangent: $\frac{dy}{dx} = 0$; vertical: $\frac{dx}{dy} = 0$

Second-degree curve:

$$x^2 \to xx_1,\ 2xy \to xy_1 + yx_1,\ 2x \to x + x_1$$

Tangent at origin: lowest-degree terms $= 0$

Even function: $f'(0) = 0$

Vertical tangent

Exists: $f'(x_0^{\pm})$ both $+\infty$ or both $-\infty$

Cusp $(+\infty, -\infty)$: none

Parametric points

Circle: $(a\cos\theta, a\sin\theta)$; ellipse: $(a\cos\theta, b\sin\theta)$

Hyperbola: $(a\sec\theta, b\tan\theta)$; $xy = c^2$:

$$\left(ct, \frac{c}{t}\right)$$

$y^2 = 4ax$: $(at^2, 2at)$; $x^2 = 4ay$: $(2at, at^2)$

$\frac{x^n}{a^n} + \frac{y^n}{b^n} = 1$:

$$\left(a\cos^{2/n}\theta,\ b\sin^{2/n}\theta\right)$$

$y = f(x)$: $(t, f(t))$; $y = mx + c$: $(t, mt + c)$

Ultimate concept

Tangent through (x_1, y_1): $f'(t) = \frac{f(t)-y_1}{t-x_1}$

Normal through (x_1, y_1):

$$-\frac{1}{f'(t)} = \frac{f(t) - y_1}{t - x_1}$$

Common tangent:

$$f'(t_1) = g'(t_2) = \frac{g(t_2) - f(t_1)}{t_2 - t_1}$$

Minimum distance

Point to curve: $m_{AP} \cdot f'(P) = -1$

Curve to line $ax + by + c = 0$: $f'(P) = -\frac{a}{b}$, $d = \frac{|ax_P + by_P + c|}{\sqrt{a^2+b^2}}$

Curve to circle: $d = CP - r$, CP normal at P

Angle, orthogonality, touching

Angle:

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

Orthogonal: $m_1 m_2 = -1$

Conics:

$$\frac{1}{a_1} - \frac{1}{b_1} = \frac{1}{a_2} - \frac{1}{b_2}$$

Touch: $f(x_1) = g(x_1)$, $f'(x_1) = g'(x_1)$

Touching values: $e^x = kx$: $k = e$; $\ln x = kx$: $k = \frac{1}{e}$; $\ln x = kx^2$: $k = \frac{1}{2e}$

Important lengths

Length of tangent:

$$\left|\frac{y_1\sqrt{1+m^2}}{m}\right|$$

Length of normal:

$$\left|y_1\sqrt{1+m^2}\right|$$

Subtangent: $\left|\frac{y_1}{m}\right|$; subnormal: $|y_1 m|$

Approximation and rates

Approximation:

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$$

Error: $f'' > 0$ under-estimate, $f'' < 0$ over-estimate

Chain:

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

Circle: $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$; sphere:

$$\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$$

Cone: $V = \frac{1}{3}\pi r^2 h$